

Bias & Failure Modes Map

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Where models break and who pays

PROFESSIONAL DEFINITION

The Bias & Failure Modes Map identifies where an AI model can produce biased, unfair or otherwise harmful outputs and who bears the consequences, guiding mitigation and monitoring.

WHO DEVELOPED IT

Drawn from algorithmic-fairness and responsible-AI research, notably the work of Cathy O'Neil and the FAccT community.

WHY IT WAS DEVELOPED

It was developed to anticipate the specific ways models fail and discriminate, since such failures carry reputational, legal and human cost.

FIGURE 1 · BIAS & FAILURE MODES MAP

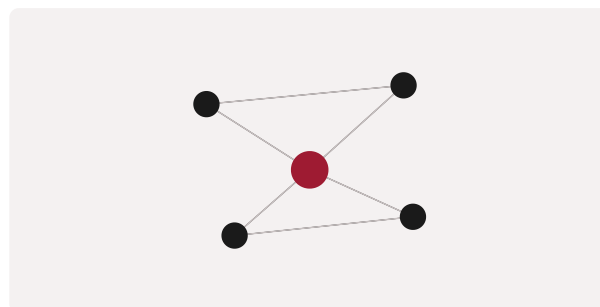


Figure 1. A schematic of the Bias & Failure Modes Map as applied in BARBM312 (illustrative).

HOW IT IS USED

Potential failure and bias modes are enumerated per use case, with detection and mitigation assigned to each.

WHEN IT IS USED

During responsible-AI design and pre-deployment review.

BY WHOM IT IS USED

Responsible-AI, risk and ML teams.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It systematises the search for harm, treating bias and failure as design concerns rather than post-hoc surprises.

RELATED FRAMEWORKS IN THIS COURSE

Cost-Benefit Trade-off Matrix · Slibrary 12

AI Risk Register · Slibrary 12

Black Swan Radar · Slibrary 12

Mitigation Cost Curve · Slibrary 12

SOURCES (HARVARD REFERENCING STYLE)

O'Neil, C. (2016) Weapons of Math Destruction. New York: Crown.

Mehrabi, N. et al. (2021) 'A survey on bias and fairness in machine learning', ACM Computing Surveys, 54(6).